

Discrete and Topological Structures

Chapter 1a: Pathak's Introduction to Nonlinear Analysis

Based on H.K. Pathak (2018)

2026

Outline

- 1 Section 1.1: Discrete Structure
- 2 Section 1.2: Topological Spaces
- 3 Summary

Partially Ordered Sets

Definition (Partial Order)

A binary relation $\preceq$ on a nonempty set X is a *partial order* if:

- P1** Reflexive: $x \preceq x$ for all $x \in X$
- P2** Antisymmetric: $x \preceq y$ and $y \preceq x$ imply $x = y$
- P3** Transitive: $x \preceq y$ and $y \preceq z$ imply $x \preceq z$

A *partially ordered set* or *poset* is a set with a partial order, denoted $(X, \preceq)$.

Examples:

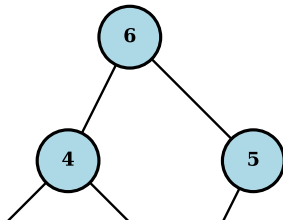
- Standard order $\leq$ on $\mathbb{R}$
- Subset relation $\subseteq$ on power sets
- Divisibility on positive integers

Chains and Extremal Elements

Definition

- A *chain* is a totally ordered subset
- An *upper bound* of S is an element u such that $s \preceq u$ for all $s \in S$
- A *least upper bound* (supremum) is the smallest upper bound
- An element is *maximal* if no larger element exists
- An element is *minimal* if no smaller element exists

Hasse Diagram of a Partially Ordered Set



Key Properties:

- Maximal elements may not be unique
- A maximum element (if it exists) is maximal
- In a total order, maximal = maximum

Zorn's Lemma

Theorem (Zorn's Lemma)

If $(X, \preceq)$ is a poset where every chain has an upper bound, then X has a maximal element.

Significance:

- Equivalent to the Axiom of Choice
- Essential for existence proofs in functional analysis
- Used in Hahn-Banach theorem, basis existence, maximal ideals

Lattices

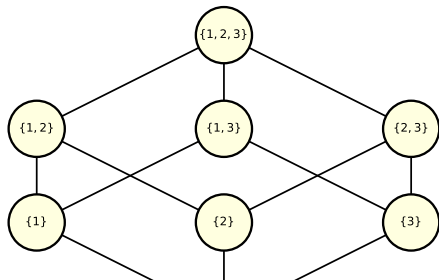
Definition (Lattice)

A *lattice* is a poset $(A, \preceq)$ where for any $a, b \in A$:

- Their *join* (least upper bound) $a \vee b$ exists
- Their *meet* (greatest lower bound) $a \wedge b$ exists

A lattice is *complete* if every subset has a supremum and infimum.

Boolean Lattice B_3 (Power Set of $\{1, 2, 3\}$)



Example: Power Set

- Elements: All subsets of $\{1, 2, 3\}$
- Join: Union
- Meet: Intersection
- Minimum: $\emptyset$
- Maximum: $\{1, 2, 3\}$

Metric Spaces

Definition (Metric)

A function $d : X \times X \rightarrow \mathbb{R}$ is a *metric* if for all $x, y, z \in X$:

M1 $d(x, y) \geq 0$ (non-negativity)

M2 $d(x, y) = 0 \iff x = y$ (identity)

M3 $d(x, y) = d(y, x)$ (symmetry)

M4 $d(x, y) \leq d(x, z) + d(z, y)$ (triangle inequality)

A *metric space* is a pair (X, d) where d is a metric.

Examples:

- Euclidean distance: $d(x, y) = |x - y|$ on $\mathbb{R}$

- Discrete metric: $d(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$

Open Sets and Continuity

Definition (Open Ball and Open Set)

An *open ball* around x with radius r is:

$$B(x, r) = \{y \in X : d(x, y) < r\}$$

A set G is *open* if for every $x \in G$, there exists $r > 0$ with $B(x, r) \subseteq G$.

Definition (Continuity)

$f : X \rightarrow Y$ is *continuous* at x_0 if for every $\epsilon > 0$, there exists $\delta > 0$ such that:

$$d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \epsilon$$



Equivalent characterization:

f is continuous iff the preimage of every open set is open.

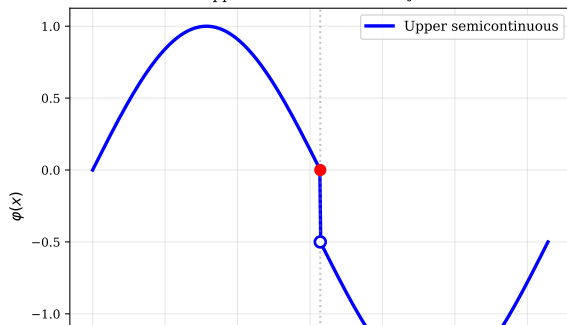
Semicontinuity

Definition (Semicontinuous Functions)

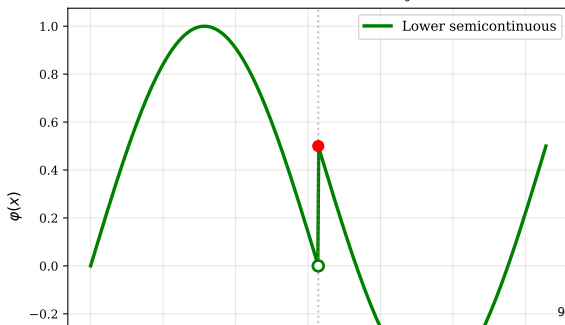
A function $\varphi : X \rightarrow \mathbb{R}$ is:

- *Upper semicontinuous* at x_0 if $\limsup_{x \rightarrow x_0} \varphi(x) \leq \varphi(x_0)$
- *Lower semicontinuous* at x_0 if $\liminf_{x \rightarrow x_0} \varphi(x) \geq \varphi(x_0)$
- *Continuous* if both conditions hold

Upper Semicontinuous at $x_0 = \pi$



Lower Semicontinuous at $x_0 = \pi$



Closed Sets and Closure

Definition (Closed Set)

A set F is *closed* if its complement $X \setminus F$ is open.

Definition (Closure)

The *closure* $\overline{C}$ of a set C is the intersection of all closed sets containing C .

Properties:

- A set is closed iff $C = \overline{C}$
- The closed ball $\overline{B}(x, r) = \{y : d(x, y) \leq r\}$ is closed
- Closure is the “smallest” closed set containing C
- Intersection of closed sets is closed

Neighborhoods and Separation Axioms

Definition (Neighborhood)

N is a *neighborhood* of x if there exists an open set G with $x \in G \subseteq N$.

T_0 Space (Kolmogorov)

For distinct $x, y \in X$:

$\exists$ open set U containing
one but not the other

(Can distinguish points)

T_1 Space

For distinct $x, y \in X$:

$\exists$ open set U_x with
 $x \in U_x, y \notin U_x$ and
 $\exists$ open set U_y with

$y \in U_y, x \notin U_y$

T_2 Space (Hausdorff)

For distinct $x, y \in X$:

$\exists$ disjoint open sets
 $U \ni x$ and $V \ni y$ with

$U \cap V = \emptyset$

Separation axioms allow distinguishing points using open sets.

Nets and Convergence

Definition (Directed Set)

$(\mathcal{D}, \geq)$ is directed if it's reflexive, transitive, and for any $\alpha, \beta \in \mathcal{D}$, there exists γ with $\gamma \geq \alpha, \beta$.

Definition (Net)

A *net* is a function from a directed set into X . The net $\{x_\alpha\}$ converges to x if for every neighborhood U of x , there exists α_0 such that $x_\alpha \in U$ for all $\alpha \geq \alpha_0$.

Topological Space X

Compactness

Definition (Compact Space)

A space X is *compact* if every open cover has a finite subcover.

Theorem (Heine-Borel)

A subset of $\mathbb{R}^n$ is compact iff it is closed and bounded.

Key properties:

- Continuous image of compact space is compact
- In compact Hausdorff spaces, limits are unique
- $[a, b]$ is compact in $\mathbb{R}$; (a, b) is not

Non-Compact Space

Compact Space

Normed Linear Spaces

Definition (Norm)

A function $\|\cdot\| : X \rightarrow [0, \infty)$ on a linear space is a *norm* if:

N1 $\|x\| = 0 \iff x = 0$

N2 $\|\lambda x\| = |\lambda| \|x\|$

N3 $\|x + y\| \leq \|x\| + \|y\|$

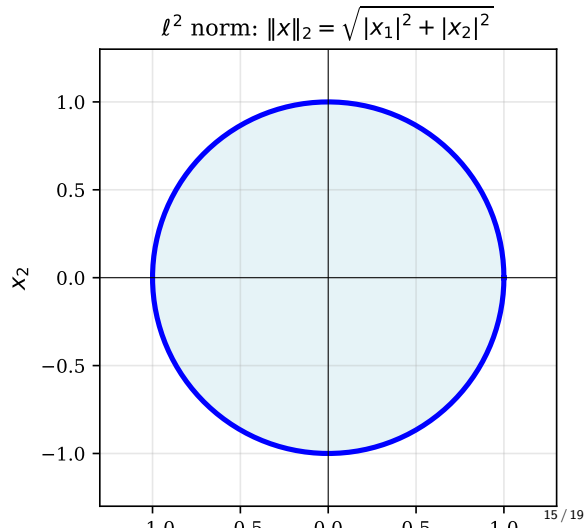
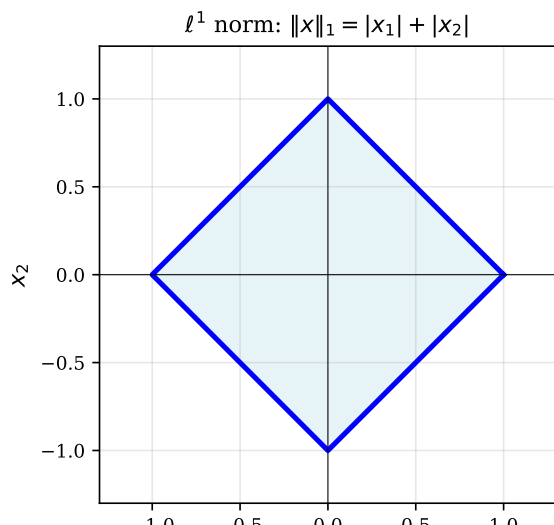
Every normed space is a metric space with $d(x, y) = \|x - y\|$.

Common norms on $\mathbb{R}^n$:

- $\|x\|_1 = \sum_i |x_i|$ (Manhattan norm)
- $\|x\|_2 = \sqrt{\sum_i |x_i|^2}$ (Euclidean norm)
- $\|x\|_\infty = \max_i |x_i|$ (Chebyshev norm)

Unit Spheres in Different Norms

The *unit sphere* is $S = \{x : \|x\| = 1\}$. Its shape depends on the norm:



Inner Product Spaces

Definition (Inner Product)

A function $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{C}$ is an *inner product* if:

IP1 $\langle x, x \rangle \geq 0$ for all x

IP2 $\langle x, x \rangle = 0 \iff x = 0$

IP3 $\langle x, y \rangle = \overline{\langle y, x \rangle}$

IP4 Linear in first argument

Every inner product induces a norm: $\|x\| = \sqrt{\langle x, x \rangle}$.

Example: On $\mathbb{R}^n$,

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

Example: The Hilbert Space ℓ^2

Definition (ℓ^2 Space)

$$\ell^2 = \left\{ (x_i)_{i=1}^{\infty} : \sum_{i=1}^{\infty} |x_i|^2 < \infty \right\}$$

With inner product: $\langle x, y \rangle = \sum_{i=1}^{\infty} x_i \overline{y_i}$.

```
import numpy as np

# Example: x = (1, 1/2, 1/4, 1/8, ...)
x = np.array([1/2**i for i in range(1, 20)])

# Compute l^2 norm
norm_x = np.linalg.norm(x)
sum_sq = np.sum(x**2)

print(f"||x||_2 = {norm_x:.6f}")
print(f"Sum x_i^2 = {sum_sq:.6f}")

# Geometric series sums to 1/3
theoretical = 1/(1 - 1/4)
print(f"Limit = {theoretical:.6f}")
```

Key Concepts

Discrete Structures (1.1):

- Partially ordered sets and chains
- Zorn's lemma for maximal elements
- Lattices with joins and meets

Topological Foundations (1.2):

- Metric spaces and distances
- Open/closed sets and continuity
- Compactness and completeness
- Normed and inner product spaces

These tools are fundamental for:

- Nonlinear functional analysis
- Fixed point theorems
- Optimization and variational methods

References

-  Pathak, H.K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer.
-  Rudin, W. (1976). *Principles of Mathematical Analysis*. McGraw-Hill.
-  Munkres, J.R. (2000). *Topology*. Prentice Hall.